Project:-

Prediction of Credit Card Default

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1. Introduction:-

This project focuses on using supervised machine learning to predict whether a credit card user is likely to default — that means they stop making payments on time.

The main goal is to help banks to find customers who are at high risk of not paying their credit card bills. If we can identify them early, banks can take steps to reduce losses.

We followed the complete in the following steps:

1. Data Cleaning and Preprocessing:-
We handled missing values, converted text-based categories into numbers, and treated unusual or extreme values.

2. Exploratory Data Analysis
We explored the dataset to find patterns and understand which factors might lead to default.

3. Feature Engineering
We created new features like the ratio of payment to bill amount, average bill amount, and sequences of missed payments to better capture financial behavior.

4. Model Building
We trained five classification models: Logistic Regression, Random Forest, Gradient Boosting, Neural Network, and XGBoost.

5. Model Evaluation
We compared the models using the F2-score, which puts more focus on correctly identifying defaulters. The model with the best F2-score was selected.

2. Data Preprocessing:-

1. Feature Selection
   We removed features with very low variance because they do not provide useful information for prediction. This was done using tools from the sklearn library.

2. Encoding Categorical Variables
   We used one-hot encoding to convert the categorical columns such as sex, marriage, and education into numerical form, which is required for machine learning models.

3. Removing Outliers
   Outliers were handled using a custom function to clip extreme values based on the interquartile range (IQR). This helps avoid the negative impact of unusual values on the model.

4. Train-Test Split
   We divided the dataset into two parts: 80% for training the model and 20% for testing it. This helps us check how well the model performs on unseen data.

5. Handling Class Imbalance with SMOTE
   The dataset had many more non-defaulters than defaulters. To fix this, we used SMOTE to create synthetic examples of defaulters in the training set. This balanced the classes and helped prevent the model from being biased toward the majority class.

6. Feature Scaling
   We standardized the numerical features using StandardScaler. This process adjusts all values so that they have a mean of 0 and a standard deviation of 1. This step is important for models like Logistic Regression and Neural Networks, which are sensitive to feature scales. The scaler was fit on the training data and then applied to both training and test sets

$$z = x - \mu / \sigma$$

3. EDA and visualization:-

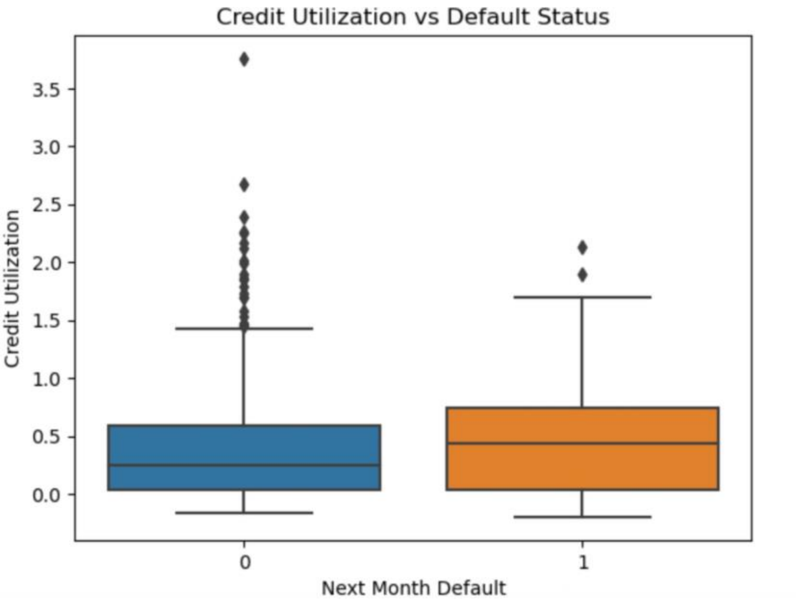
Feature Engineering:-

To improve model performance, we created two new features:

(1) Credit utilization:

Formula: AVG_Bill_amt divided by LIMIT_BAL

People who defaulted on their payments generally used a larger portion of their available credit limit. The median credit utilization is higher for defaulters, indicating that, on average, they tended to use more credit compared to non-defaulters. Additionally, defaulters are more likely to have extreme cases where they used very high amounts of credit. In contrast, most non-defaulters kept their credit usage relatively low. These observations suggest that higher credit utilization is associated with an increased likelihood of default.

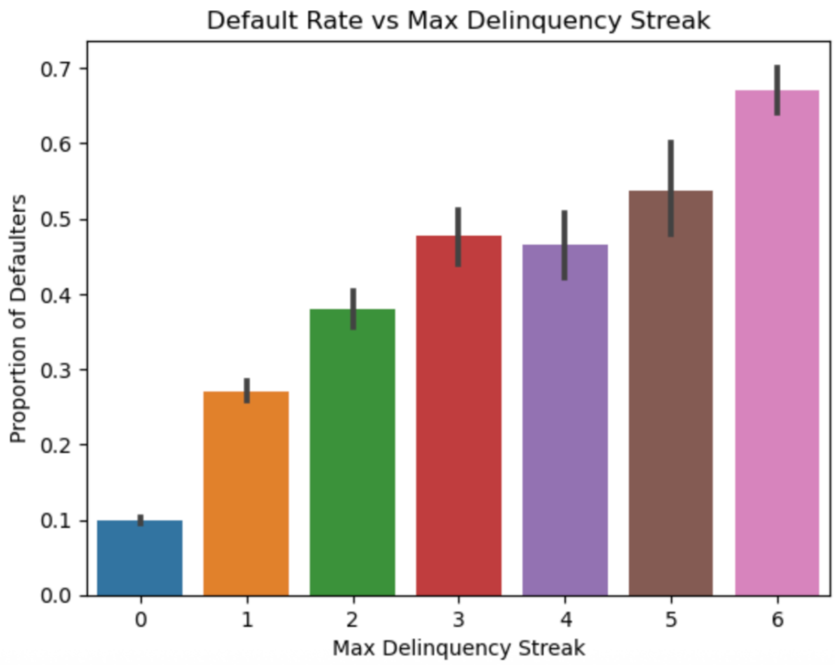


(2) Max delinquency streak:-

This feature tracks the maximum number of consecutive months in which a customer made late payments. It helps identify long-term payment issues and is a strong signal for default risk.

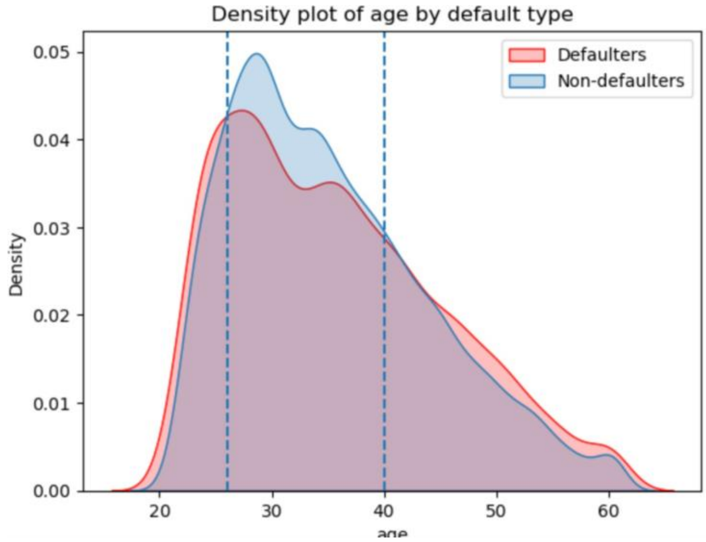
- Customers with no delinquency have a very low chance of default (around 10%).
- As the delinquency streak increases, the proportion of defaulters also increases.
- When the streak reaches 6 months the default rate goes above **65%**.

This pattern shows that customers who regularly miss payments for several months are much more likely to default So, Max delinquency streak is an important feature for predicting credit card default.



(3) Age Distribution by Class:-

Based on visual analysis, we observed that individuals between the ages of 25 and 40 are less likely to default. We can use this insight to create an age-based feature, such as grouping customers into risk categories (e.g., below 25, 25 to 40, above 40), to help the model better capture patterns in repayment behaviour.

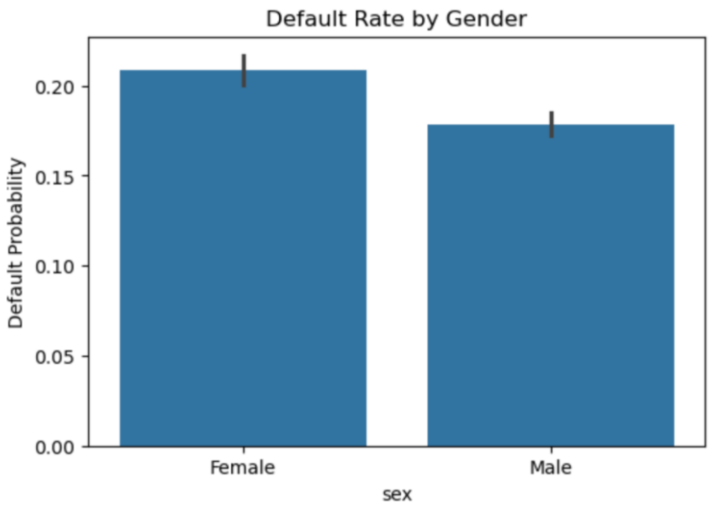


Default rate based on gender:-

Females have a slightly higher default rate compared to males.

The default probability for females is above 0.20, while for males it is below 0.20.

This suggests that gender might play a small role in credit default behavior, and could be considered as one of the predictive features in credit risk models.



4. Financial Analysis of Default Drivers:-

Several money habits can show if a person is at high risk of not paying back their credit:

1. **Late Payments (pay_i):**
   If a person pays their bills late many times or for many months, it is a clear sign that they might not pay back in the future. The more often they are late, the higher the risk of default.

2. **Credit Utilization:**
   This shows how much of their credit limit the person is using. If someone uses 80% or more of their available credit, it often means they are under financial pressure. This increases the chances that they will not be able to repay the loan.

3. **Repayment History:**
   Looking at how much they usually pay and if they've missed payments in a row helps us understand their behavior. People who regularly pay less or skip payments are more likely to default.

4. **Gender(sex):**Females show a slightly higher default rate than males. This may suggest that women in the dataset face slightly more financial stress or challenges in repayment. However, the difference is small and should be considered with caution.

5. Model Comparison:-

We used different supervised machine learning models to build a system that can predict if a person will default on their credit card payment next month. These models are based on mathematical concepts and were tested on our dataset.

The five models we used are:

1. Logistic Regression
2. Random Forest (a tree-based ensemble method)
3. MLPClassifier (a type of neural network)
4. XGBoost (another powerful tree-based method)

Model	F2 Score	Accuracy	F1 Score	Threshold
Logistic Regression	0.567	0.463	0.376	0.25
Random Forest	0.613	0.584	0.434	0.25
MLPClassifier	0.591	0.505	0.399	0.15
XGBoost	0.593	0.553	0.414	0.10

Final Model Chosen:

We selected the Random Forest model because it gave the highest F2 score. This means it was the best at correctly finding defaulters while keeping the number of missed defaulters (false negatives) low.

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Results for Random forest
Optimal Threshold: 0.25
Accuracy: 0.584
F1 Score: 0.434
F2 Score: 0.613
Confusion Matrix:
[[2145 1950]
 [ 149  806]]
Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.52      0.67      4095
           1       0.29      0.84      0.43       955

    accuracy                           0.58      5050
   macro avg       0.61      0.68      0.55      5050
weighted avg       0.81      0.58      0.63      5050
```

6. Business Implications:-

Machine learning models help businesses make safer, faster, and smarter decisions. Here's how:

Behavioral Risk Forecasting:
the model can find customers who might miss their payments before it actually happens.
It does this by looking at their past financial behavior and payment history.
By knowing this early, banks can act in time and reduce the risk of full loan default, which is harder and more expensive to deal with.
This also helps banks keep their loan portfolio healthy.

Policy Design:
The results from the model can help businesses create or update their policies.
For example, features like age, education level,gender, repayment history, and number of delayed payments (delinquency streak) can guide banks in deciding who to give credit to and under what conditions.
So, the business can set better rules based on data, not just guesswork.

Operational Efficiency:
These models can automatically detect future risks, which saves time and reduces manual work.
Earlier, checking customer risk took a lot of time and effort from bank staff.
Now, decisions can be made quickly with less human effort, saving both time and money.

7 Summary and Key Learnings:-

Strong predictive features:

The most useful features for predicting credit card default were the payment delay features (`pay_1` to `pay_6`). These show whether a person has paid late in the past. If someone regularly misses payments, it is more likely they will default in the future.

Credit utilization (how much of the credit limit is used) is also important. If someone uses a large portion of their credit, it can mean they are financially stressed.

Another useful feature was max_deliquency_streak which tells us the longest period of back-to-back months with late payments. People with long streaks of missed payments are at higher risk of defaulting.

Addressing Class Imbalance Effectively:

In our data, there were many more non-defaulters than defaulters. This imbalance makes it hard for models to learn from the minority (defaulter) class. To fix this, we used SMOTE, a method that creates new, similar examples of defaulters. This helped balance the data and improved the model's ability to recognize defaulters. Because of this, recall (the model’s ability to catch defaulters) improved a lot.

Emphasis on F2-Score for Evaluation:

In finance, it's more important to find defaulters (recall) than to avoid false alarms (precision). If the model misses a defaulter, the bank can lose money. That’s why we used the F2-score, which gives more importance to recall than precision. We also adjusted the decision threshold for each model to increase the F2-score and catch more defaulters.

Best Model:

Among all models, the Random Forest performed the best. With some parameter tuning and a threshold of 0.25, it gave an F2-score close to 0.60. This model focused on identifying more defaulters, which helps reduce financial losses for banks.